

Multi-Resolution Image Blending Using Gaussian and Laplacian Pyramids

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Abstract—Multi-resolution image blending is a widely used technique in digital image processing and computer vision for combining multiple images into a single, visually realistic result. This project implements image blending using Gaussian and Laplacian pyramids to achieve smooth transitions between images while preserving edges and fine details. Instead of directly merging images at the pixel level, the method operates across different scales. Gaussian pyramids represent images at multiple resolutions by repeated smoothing and downsampling, while Laplacian pyramids capture edge and detail information at each level. By applying a gradient mask and performing pyramid-based blending, the system reduces visible seams and produces natural-looking composites. Results demonstrate that multi-scale blending outperforms basic techniques in visual quality and artifact reduction.

Index Terms—Image blending, Gaussian pyramid, Laplacian pyramid, multi-resolution, image compositing, digital image processing

I. INTRODUCTION

Multi-resolution image blending is a widely used technique in digital image processing and computer vision for combining multiple images into a single, visually realistic result. Instead of directly merging images at the pixel level, this method operates across different scales using Gaussian and Laplacian pyramids. This multi-scale approach allows smooth regions and fine details to be handled separately, which improves overall visual quality.

In simple cut-and-paste editing, images often show visible seams, sharp boundaries, or mismatched colors. Multi-resolution blending reduces these problems by gradually combining images from low-resolution (coarse) to high-resolution (fine) levels. As a result, transitions become smooth while important structures like edges and textures are preserved. This makes the final composite look natural and consistent.

II. OBJECTIVE OF THE PROJECT

The primary objectives of this project are as follows:

- Implement image blending using Gaussian and Laplacian pyramids.
- Produce smooth transitions between two images.
- Preserve edges and fine details during blending.
- Understand the concept of multi-scale image representation.

III. PROBLEM STATEMENT

Image blending combines multiple images into a single output, but basic methods like averaging or simple masking often produce visible seams, poor transitions, and loss of important details. These problems become more noticeable when images have differences in lighting, color, or texture.

The main challenge is to create a smooth and natural blend while preserving both overall appearance and fine details. Traditional techniques fail to handle these aspects effectively.

This project addresses the problem by using multi-resolution image blending with Gaussian and Laplacian pyramids to achieve seamless transitions and better visual quality.

IV. IMPLEMENTATION

The implementation of this project is based on multi-resolution image blending using Gaussian and Laplacian pyramids.

First, two input images are loaded and resized to the same dimensions so that they can be processed together. After that, a mask is created to define which part of each image will be used in the final blending. The mask can be a simple binary or gradient mask to ensure smooth transition.

Next, Gaussian pyramids are generated for both input images. These pyramids represent the images at multiple

resolutions by repeatedly smoothing and downsampling. Then, Laplacian pyramids are constructed from the Gaussian pyramids to capture edge and detail information at each level.

After building the pyramids, the blending process is performed. At each level of the Laplacian pyramid, the corresponding layers of both images are combined using the mask. This ensures that blending occurs gradually across different scales.

Finally, the blended image is reconstructed by expanding and combining the pyramid levels step by step. The result is a smooth and natural-looking image with reduced seams and preserved details.

V. PROJECT WORKFLOW

A. Step 1: Load Input Images

```
img1_original = load_image_rgb(IMG1_PATH)
img2_original = load_image_rgb(IMG2_PATH)

print("Image A Shape:", img1_original.shape)
print("Image B Shape:", img2_original.shape)
```

Image A Shape: (1000, 1000, 3)
Image B Shape: (626, 594, 3)

Fig. 1. Loading input images for blending.

The input images are loaded into the system using image processing functions. These images serve as the primary data for the blending operation and must be of compatible size and format for accurate processing.

B. Step 2: Display Original Images

```
fig, axes = plt.subplots(1, 2, figsize=(14, 6))

axes[0].imshow(img1_original)
axes[0].set_title("Original Image A")
axes[0].axis("off")

axes[1].imshow(img2_original)
axes[1].set_title("Original Image B")
axes[1].axis("off")

plt.tight_layout()
plt.show()
```

Fig. 2. Display of original input images before processing.

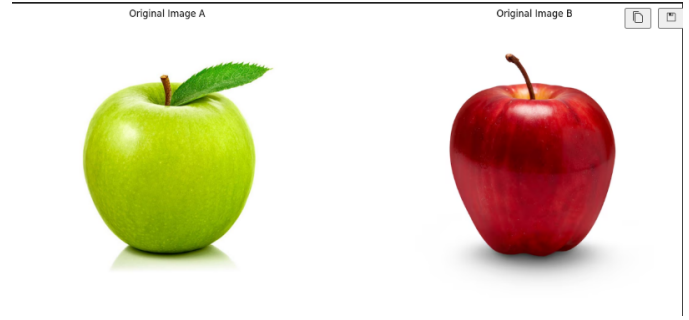


Fig. 3. Original input images showing differences in color and structure.

The original input images are displayed before any processing. This step helps to observe differences in color, lighting, and structure, which influence the blending quality.

C. Step 3: Define Gaussian Pyramid

```
def build_gaussian_pyramid(image, levels=6):
    pyramid = [image]
    current = image.copy()

    for _ in range(levels - 1):
        current = cv2.pyrDown(current)
        pyramid.append(current)

    return pyramid
```

Fig. 4. Gaussian pyramid function definition.

A function is defined to construct the Gaussian pyramid. This pyramid represents the image at multiple resolutions by repeatedly smoothing and downsampling the image.

D. Step 4: Define Laplacian Pyramid Function

```
def build_laplacian_pyramid(gaussian_pyramid):
    laplacian_pyramid = []

    for i in range(len(gaussian_pyramid) - 1):
        current = gaussian_pyramid[i]
        next_level = gaussian_pyramid[i + 1]

        expanded = cv2.pyrUp(next_level, dstsize=(current.shape[1], current.shape[0]))
        laplacian = current - expanded
        laplacian_pyramid.append(laplacian)

    laplacian_pyramid.append(gaussian_pyramid[-1])
    return laplacian_pyramid
```

Fig. 5. Laplacian pyramid function definition.

This step defines the Laplacian pyramid function, which extracts edge and texture details by subtracting consecutive levels of the Gaussian pyramid.

E. Step 5: Build and Visualize Gaussian Pyramids

```
gp_img1 = build_gaussian_pyramid(img1, LEVELS)
gp_img2 = build_gaussian_pyramid(img2, LEVELS)
gp_mask = build_gaussian_pyramid(mask, LEVELS)

print("Gaussian pyramids created successfully.")
```

Fig. 6. Building and visualizing Gaussian pyramids.



Fig. 7. Gaussian pyramid levels showing progressively lower resolution.

The Gaussian pyramid is visualized at different levels. Each level shows a lower-resolution and more blurred version of the image, capturing the overall structure and intensity variations.

F. Step 6: Build and Visualize Laplacian Pyramids

```
lp_img1 = build_laplacian_pyramid(gp_img1)
lp_img2 = build_laplacian_pyramid(gp_img2)

print("Laplacian pyramids created successfully.")
```

Laplacian pyramids created successfully.

Fig. 8. Building and visualizing Laplacian pyramids.

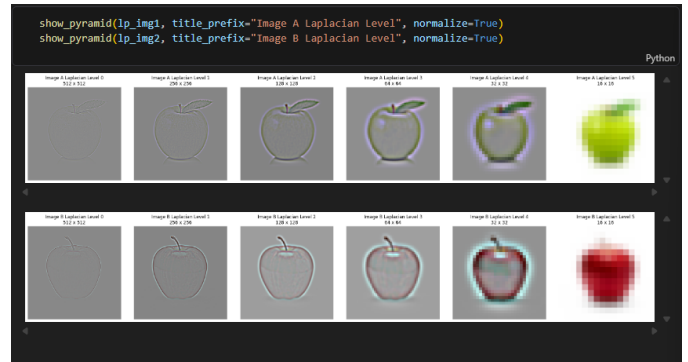


Fig. 9. Laplacian pyramid highlighting edges and fine details at each level.

The Laplacian pyramid is visualized to highlight fine details such as edges and textures. These details are essential for preserving image sharpness during blending.

G. Step 7: Mask Creation

```
plt.figure(figsize=(10, 4))
plt.imshow(mask[:, :, 0], cmap="gray")
plt.title("Blending Mask")
plt.axis("off")
plt.show()
```

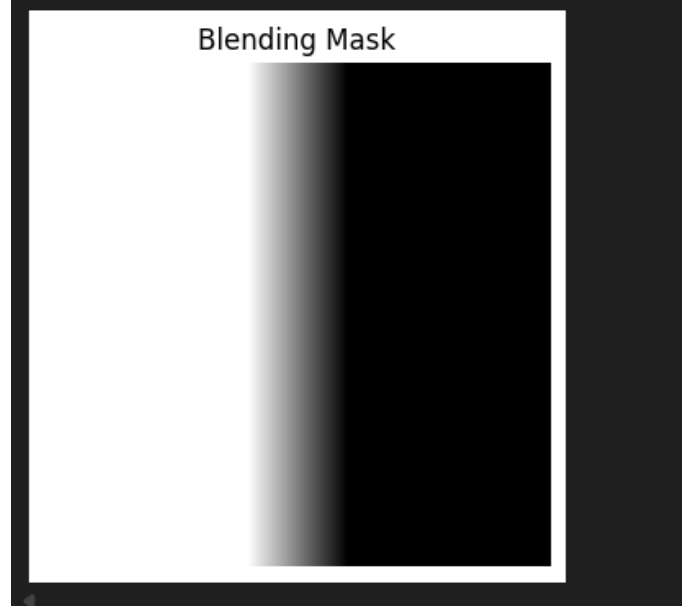


Fig. 10. Gradient blending mask for smooth transition control.

23. Blend Pyramid Levels

```
blended_pyramid = []
for la, lb, gm in zip(lp_img1, lp_img2, gp_mask):
    blended_level = gm * la + (1.0 - gm) * lb
    blended_pyramid.append(blended_level)
print("Pyramid blending completed.")
```

Fig. 11. Mask pyramid applied at multiple resolution levels.

In this step, a blending mask is created to control how the two images are combined. The mask defines which portions of each image will be visible in the final output. Instead of using a sharp boundary, a smooth gradient mask is applied to ensure a gradual transition between the images.

This smooth transition helps to avoid visible seams and sudden changes in intensity or color. The mask is also converted into a Gaussian pyramid so that it can be applied consistently at different resolution levels during the blending process. Proper mask design plays a crucial role in achieving a natural and visually pleasing final result.

H. Step 8: Blending Process

At each level of the pyramid, the Laplacian images from both inputs are combined using the mask. This multi-resolution blending ensures that both global and local features are merged smoothly.

I. Step 9: Visualize Blended Pyramid



Fig. 12. Blended pyramid showing intermediate merged results at different scales.

The blended pyramid shows intermediate results where both images are merged at different scales. This step demonstrates how blending occurs progressively across resolutions.

J. Step 10: Final Output Image

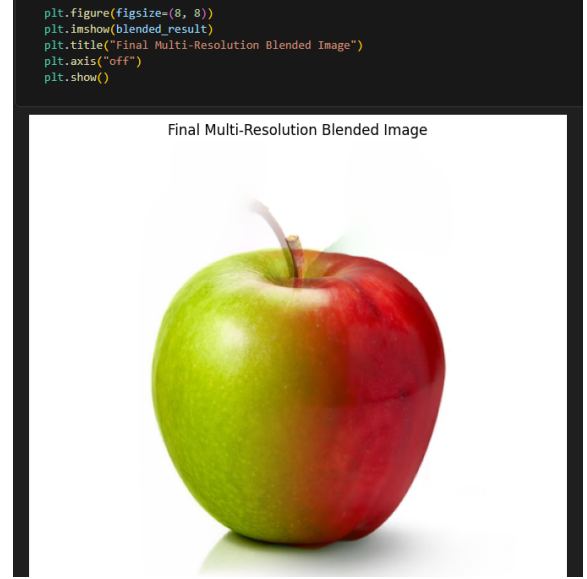


Fig. 13. Final reconstructed blended image with seamless transitions.

The final blended image is reconstructed from the combined Laplacian pyramid by successively expanding and adding each level. This process restores both the overall structure and fine details of the image. The result is a visually smooth and natural composite where transitions between the input images are seamless, with minimal visible seams or artifacts. Differences in color, lighting, and texture are effectively balanced, producing a high-quality blended output.

VI. RESULTS AND ANALYSIS

The proposed multi-resolution image blending system produced visually smooth and seamless composite images. The blending results showed minimal visible seams, even in regions where the input images had noticeable differences in color and intensity.

By using Gaussian and Laplacian pyramids, the system was able to handle both low-frequency information (such as overall shading and color) and high-frequency details (such as edges and textures). This resulted in better detail preservation and more natural-looking outputs compared to basic blending methods like direct averaging.

The use of a gradient mask further improved the transition between images by avoiding abrupt changes. However, it was observed that the quality of the final output depends on proper image alignment and an appropriate mask design.

Overall, the method achieved effective blending with improved visual quality and reduced artifacts.

VII. IMPORTANCE OF THE PROJECT

Multi-resolution image blending is an important technique in digital image processing as it improves the quality of

image compositing. Unlike simple blending methods, it ensures smooth transitions between images while preserving both overall appearance and fine details.

This project helps in understanding key concepts such as Gaussian and Laplacian pyramids, multi-scale representation, and frequency-based image processing. These concepts are widely used in advanced computer vision tasks.

In addition, the technique reduces visible seams and artifacts, making the final output more realistic and visually appealing. Therefore, this project is valuable both for academic learning and practical applications in modern imaging systems.

VIII. REAL-LIFE APPLICATIONS

- **Photo Editing and Graphic Design:** Used to combine multiple images into a single composition with smooth transitions. It helps designers create realistic visuals without visible seams or harsh boundaries.
- **Panorama Stitching:** Multiple overlapping images are blended to create wide-angle or 360-degree panoramic views. Multi-resolution blending ensures seamless merging even when lighting and colors differ.
- **Film and Visual Effects (VFX):** Widely used in movies to merge real-world footage with computer-generated imagery (CGI). It helps produce high-quality scenes where transitions appear natural and visually consistent.
- **Mobile Photography (HDR & Portrait Mode):** Smartphones use similar blending techniques to combine multiple exposures in HDR images and to enhance portrait effects with better background separation and smooth edges.
- **Medical Imaging:** Different types of medical scans (such as MRI, CT) can be blended to provide more detailed and informative visuals for diagnosis and analysis.
- **Satellite Imaging and Remote Sensing:** Used to combine multiple satellite images into a large, continuous map. This helps in geographical analysis, weather monitoring, and environmental studies.
- **Augmented Reality (AR):** Helps in placing virtual objects into real-world environments by blending them naturally with the background, improving realism and user experience.
- **Surveillance and Security Systems:** Multiple camera feeds can be combined to provide a wider and clearer view, improving monitoring and analysis.

IX. CONCLUSION

This project successfully demonstrates the implementation of multi-resolution image blending using Gaussian and Laplacian pyramids. The method effectively combines two images by processing them at different resolution levels, resulting in smooth transitions and preserved details.

The results show that multi-scale blending produces significantly better outputs compared to basic blending techniques. It maintains both global consistency and local features such as edges and textures.

Overall, the project highlights the effectiveness of pyramid-based techniques in image processing and provides a strong

foundation for further exploration in computer vision and advanced image manipulation.

X. FUTURE IMPROVEMENTS

- Use more advanced and adaptive masks to improve blending quality in complex regions.
- Extend the system to blend multiple images instead of only two.
- Improve image alignment techniques to reduce distortion and mismatching.
- Optimize the algorithm for faster performance and real-time processing.
- Apply the method to video blending for smooth frame transitions.
- Integrate deep learning techniques to automate mask generation and improve results.

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